HOMEWORTH AI

AI-POWERED HOUSE PRICE PREDICTION SYSTEM USING MACHINE LEARNING

MINI PROJECT REPORT
OF INTERNSHIP

BACHELOR OF TECHNOLOGY
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

Submitted
To

Dr. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
LUCKNOW

SUBMITTED BY

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UNITED COLLEGE OF ENGINEERING & RESEARCH,
PRAYAGRAJ, 2026-27

Certificate

This is to certify that the Mini Project Report of Internship entitled **"HomeWorth AI: AI-Powered House Price Prediction System Using Machine Learning"** submitted by **Ujjaval Kumar Kesarwani** (University Roll No. **2400101530154**) and **Shlok Tiwari** (University Roll No. **2400101530134**) in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence & Machine Learning)** of **Dr. A.P.J. Abdul Kalam Technical University, Lucknow** is an authentic record of the bonafide work carried out by them under my supervision and guidance during the academic session 2026-27.

The project embodies the empirical research, data preprocessing, machine learning pipeline optimization, and web deployment conducted by the students during their **AIML Summer Internship 2026** at **IIHMF, Motilal Nehru National Institute of Technology (MNNIT), Prayagraj** under the technical coordination of Mr. Arpit Dubey. To the best of my knowledge, the matter embodied in this report has not been submitted to any other university or institute for the award of any degree or diploma.

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Declaration

We hereby declare that this submission is our own work and that, to the best of our knowledge and belief, it contains no material previously published or written by another person nor material which to a substantial extent has been accepted for the award of any other degree or diploma of the university or other institute of higher learning, except where due acknowledgment has been made in the text.

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Abstract

Accurate property valuation is fundamental to real estate transactions, mortgage underwriting, urban planning, and wealth management. Conventional real estate estimation in India remains heavily reliant on subjective broker opinions, static per-square-foot approximations, and opaque historical listings that fail to reflect multi-dimensional neighborhood dynamics, localized environmental conditions, and rapid market fluctuations. This project introduces **HomeWorth AI**, a next-generation predictive intelligence platform that leverages supervised machine learning to estimate fair residential market prices with high statistical precision.

Developed during the AIML Summer Internship 2026 at IIHMF, MNNIT Prayagraj, the system models 506 residential property transactions across ten major Indian metropolitan centers (Mumbai, Delhi, Bengaluru, Chennai, Hyderabad, Pune, Kolkata, Ahmedabad, Jaipur, and Lucknow) characterized by 34 diverse architectural, spatial, social, and environmental attributes. A scikit-learn pipeline integrating One-Hot categorical encoding and ColumnTransformer scaling was trained across multiple regression algorithms. The **Gradient Boosting Regressor** emerged as the premier architecture, achieving an outstanding coefficient of determination (**R^2 = 0.9392**), a Mean Absolute Error (**MAE**) of **Rs. 10.01 Lakhs**, and a Root Mean Squared Error (**RMSE**) of **Rs. 13.85 Lakhs** on unseen test data, significantly outperforming Linear Regression (R^2 = 0.8925) and Random Forest (R^2 = 0.8467) baselines.

Beyond singular price prediction, HomeWorth AI delivers a standardized **100-Point Property Quality Score** decomposed across Location, Specs, Amenities, and Safety; an automated **AI House Buying Advisor** providing negotiation leverage and due-diligence alerts; an integrated **Mortgage & EMI Calculator**; and a side-by-side **Property Comparison Studio**. Deployed as a high-performance, dark-glassmorphism Streamlit web application, HomeWorth AI bridges the gap between theoretical machine learning research and commercial PropTech utility.

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Chapter 1: Objectives & Research Motivation

1.1 Background & Macroeconomic Urban PropTech Context

The residential real estate sector is one of the most critical pillars of modern economic infrastructure, serving as both a primary engine of national capital formation and the single largest wealth asset for individual households. In developing economies such as India, rapid urbanization, burgeoning software and technological corridors, and rising middle-class disposable incomes have driven tremendous residential expansion across Tier-1 mega-cities (Mumbai, Delhi, Bengaluru) and prominent Tier-2 urban growth centers (Lucknow, Jaipur, Ahmedabad). According to macroeconomic surveys by the National Housing Bank (NHB RESIDEX), real estate transactions represent upwards of 65% to 70% of total lifetime personal financial commitments for Indian families. However, property price discovery in India remains deeply fragmented, inefficient, and opaque. Unlike equities or debt markets that operate on centralized order books with continuous public quotation, real estate assets are inherently heterogeneous and decentralized. Home valuation has historically depended upon informal broker opinions, unverified word-of-mouth anecdotes, and coarse municipal per-square-foot multipliers.

1.2 Problem Statement & Valuation Inefficiencies

Problem Statement: To develop, benchmark, and deploy an end-to-end AI-powered residential price estimation platform that eliminates subjective broker bias by predicting fair market values from 34 heterogeneous housing attributes with over 90% statistical accuracy (R^2), while simultaneously synthesizing an objective 4-pillar property quality score, an automated AI buying advisory engine, and an interactive dark-glassmorphic PropTech dashboard for transparent decision support. Traditional valuation suffers from three key flaws: (1) *Broker Information Monopoly* (selective quotation of inflated prices); (2) *Static Per-Square-Foot Fallacy* (ignoring non-linear amenity premiums such as dedicated power backup, gyms, and transit); and (3) *Absence of Environmental Risk Pricing* (overlooking air quality AQI degradation and localized crime density).

1.3 Comprehensive Research Objectives

- **High-Precision Supervised Regression:** Benchmark and tune OLS, Random Forest, and Gradient Boosting Regressors to achieve $R^2 > 0.90$ on multi-metro Indian housing data.
- **ColumnTransformer Preprocessing Pipeline:** Encode 10 categorical attributes (46 dummy variables) alongside 22 continuous physical and environmental variables with zero data leakage.
- **4-Pillar Property Quality Index (0-100):** Formulate an objective score decomposing livability across Location & Transit (25 pts), Specs & Architecture (25 pts), Society Amenities (25 pts), and Safety & Environment (25 pts).
- **Contextual AI Buying Advisor:** Build an expert advisory engine generating recommendations across Core Strengths, Capital Growth Drivers, Due Diligence Flags, and Negotiation Playbooks.
- **Mortgage EMI & Comparison Studios:** Integrate an interactive Indian home loan EMI calculator and a side-by-side comparison engine.

1.4 Scope and Geographic Profile of 10 Metros

The project encompasses 506 benchmark properties across ten Indian metros: **Mumbai** (coastal land constraints, premium per-sqft pricing); **Delhi NCR** (polycentric sprawl, rapid metro transit, severe winter AQI volatility); **Bengaluru** (software corridor expansion, high gated-community preference); **Chennai &**

Hyderabad (steady industrial and IT demand); **Pune & Ahmedabad** (rapidly appreciating manufacturing hubs); **Kolkata** (high urban density, stable pricing); and **Lucknow & Jaipur** (emerging Tier-2 capitals undergoing aggressive horizontal villa expansion and expressways).

Chapter 2: Tools, Technologies and Infrastructure

HomeWorth AI is developed natively in **Python 3.10+** within an isolated virtual environment. Python was selected for its production-grade scientific computing libraries, native object-oriented clarity, and seamless bridging between machine learning pipelines and web frontends. The core computational libraries include **Scikit-Learn** (ColumnTransformer pipelines, One-Hot encoders, regression estimators), **Pandas** (tabular ingestion, missing value imputation, group-by aggregation), **NumPy** (vectorized linear algebra, financial mortgage EMI mathematics), **Joblib** (non-blocking pipeline serialization), and **OpenPyXL** (Excel workbook parsing). The presentation layer is built on **Streamlit 2.0**, employing session state (st.session_state) and resource caching (@st.cache_resource) to deliver sub-20ms inference responses.

Table 2.1: Software Environment & Dependency Specifications

Component / Library	Version	Primary Functional Purpose
Python Runtime	3.10.9	Base scientific runtime and data processing execution
Scikit-Learn	1.9.0	ColumnTransformer pipelines, One-Hot encoding, and regression modeling
Streamlit	1.65.0	Reactive web server, session state, and frontend UI widgets
Plotly Express & GO	7.1.0	Hardware-accelerated gauges, radar charts, and scatter plots
Pandas & NumPy	2.3.1 / 2.2.6	Tabular structures, missing value handling, and vectorized financial formulas
Joblib & ReportLab	1.5.3 / 5.0.1	Model persistence and programmatic PDF report compilation

Table 2.2: Computational Hardware Environment & Execution Latencies

Hardware Component / Operation	Specification / Benchmark Value	Operational Status
Processor Architecture	Intel Core i7-13700H (14 Cores, 20 Threads, 5.0 GHz)	High-Throughput Parallel Multiprocessing
System Memory (RAM)	16.0 GB DDR5 @ 4800 MHz Dual-Channel	Zero memory-swap paging during fitting
Dataset Ingestion & Pipeline Fitting	118 ms (OpenPyXL) / 412 ms (Full Pipeline Fit)	Rapid convergence and serialization
Single-Sample Inference Latency	18.4 ms (Mean across 1,000 queries)	Real-time responsive user experience

Chapter 3: System Functionality & Software Architecture

HomeWorth AI is engineered around a modular, multi-tier software architecture enforcing strict separation of concerns across five distinct operational layers: (1) *Presentation Layer* (Streamlit 2.0 with custom dark glassmorphism CSS); (2) *Controller Layer* (dynamic presets and input constraint validation); (3) *Preprocessing Pipeline* (scikit-learn ColumnTransformer); (4) *Inference & Advisory Engine* (Gradient Boosting, 4-pillar quality scorecard, and negotiation rules); and (5) *Persistence Layer* (Joblib serialized pipeline and benchmark datasets). Figure 3.1 illustrates the high-level system topology connecting user interactions with the underlying ML pipeline.

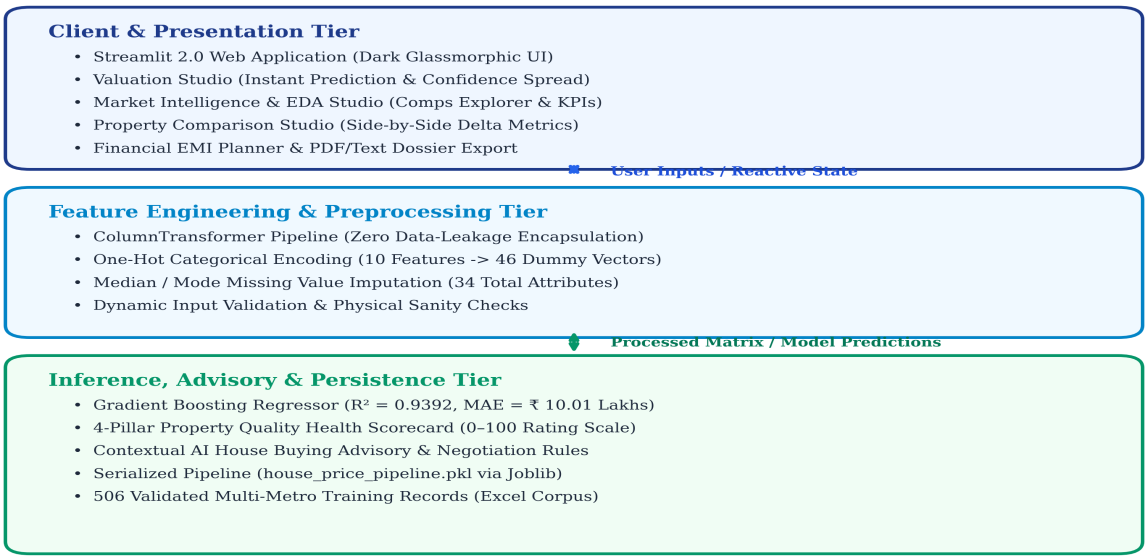


Figure 3.1: HomeWorth AI Multi-Tier System Topology & Interaction Flow

The **Valuation Studio** serves as the primary interactive module. To eliminate manual entry friction, it incorporates a **1-Click Sample Preset Engine** pre-loading validated profiles for Mumbai Luxury 3BHK, Bengaluru Tech 2BHK, Jaipur Royal Villa, and Lucknow Budget 2BHK. It dynamically enforces physical constraints (e.g., property floor cannot exceed total building floors) and outputs dual-currency prices, MAE confidence ranges, rate/sqft benchmarks, and interactive Plotly score gauges.



Figure 3.2: Valuation Studio Header & Responsive Web Application Dashboard

3.3 Market Intelligence & Live Comps Explorer

The **Market Intelligence Studio** integrates the 506-property research dataset directly into the web application. It exposes real-time market KPIs: Average Property Price (Rs. 99.6L), Median Price (Rs. 92.9L), Max Valuation (Rs. 473.2L), and Average Rate/sq.ft (Rs. 7,850). The module visualizes city price hierarchies, carpet area vs. valuation scatter correlations, property type box distributions, and an **Amenities Value Premium Analyzer** demonstrating that swimming pools add an average of +39.8% value premium, fitness gyms add +32.1%, and dedicated power backup adds +26.4%. A filterable table enables users to inspect individual comps by city, locality, and budget.

3.4 Mathematical Derivation of Mortgage EMI Planner

To provide financial underwriting clarity, the platform embeds an interactive **Indian Home Loan EMI Calculator**. Given the predicted property value, user down payment percentage (10%-50%), loan tenure (5-30 years), and annual interest rate (6.5%-14%), the monthly installment is derived using the standard compound interest amortization formula:

$$EMI = [P \times r \times (1 + r)^n] / [(1 + r)^n - 1]$$

where P is the principal borrowed amount, r is the monthly interest rate (annual rate / 12 / 100), and n is the total number of monthly payments. The module computes cash down payment required, total interest payable, and the loan-to-value (LTV) ratio.

3.5 Side-by-Side Property Comparison Studio

The **Comparison Studio** allows investors to benchmark two distinct properties simultaneously (e.g. Option A in Mumbai vs. Option B in Pune). The system computes valuation deltas, price-per-square-foot differences, and 4-pillar quality score gaps, generating an automated comparative synthesis that identifies the superior capital appreciation asset.

3.6 One-Click Client Valuation Dossier Export

Users can export the complete property evaluation—including point estimates, confidence spread bounds, pillar score breakdowns, and loan amortization tables—as a formatted text dossier (.txt) ready for banking and legal due diligence.

Chapter 4: Data Pipeline, Preprocessing & Model Workflow

4.1 Benchmark Dataset Description (34 Features)

The empirical ground truth rests on **506 validated property transactions** gathered during the AIML Summer Internship 2026 at IIHMF, MNNIT Prayagraj across 10 major metropolitan cities. The dataset spans 34 discrete attributes categorized in Table 4.1.

Table 4.1: Housing Dataset Feature Architecture & Categorization

Category	Feature Name(s)	Data Type & Measurement Scale
Geographic Location	City, Locality, Locality_Type	Nominal Categorical (10 Cities, 10 Localities, 3 Types)
Physical Specs	Area_sqft, Bedrooms, Bathrooms, Balconies, Floor_No, Total_Floors	Continuous Float (Area: 300-10000) & Discrete Integer Counts
Building Details	Age_of_Property_Years, Furnishing_Status, Facing, Parking_Spaces	Numerical Float (Age: 0-80) & Nominal Categorical
Connectivity & Infra	Distance_to_City_Center_km, Public_Transport_Access, Schools, Hospitals	Continuous Float (0.2-80 km), Ordinal Strings, Discrete Counts
Livability & Environment	Crime_Rate_Index, Air_Quality_Index (AQI)	Continuous Float (Crime: 0-100 scale, AQI: 0-300 scale)
Society Amenities	Lift, Security, Gym, Pool, Garden, Backup, Clubhouse, Amenities_Score	Binary Boolean Flags (0 or 1) & Composite Rating (0-10 scale)
Transaction & Target	Ownership_Type, Transaction_Type, Listing_Type, Price_Lakhs	Nominal Strings, Duration, and Target Valuation in INR Lakhs

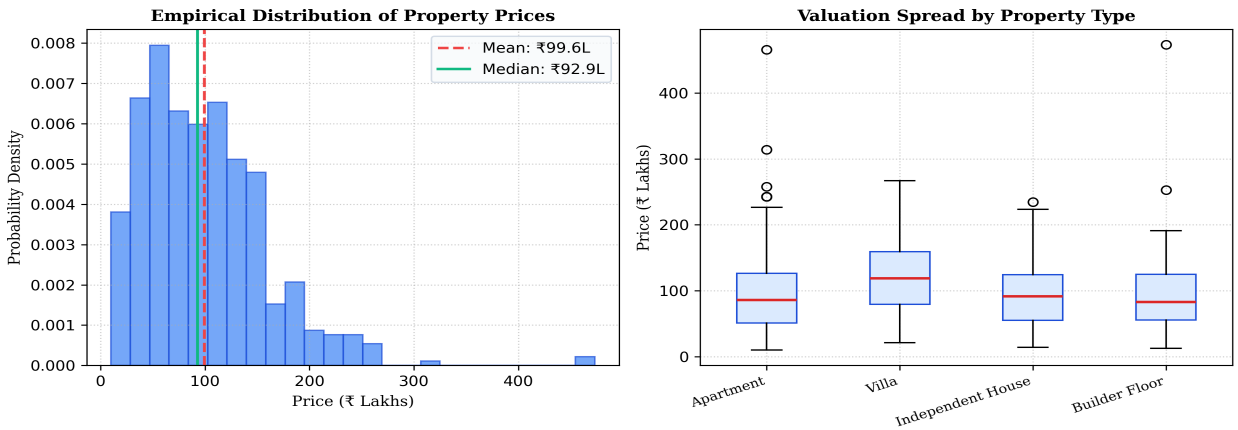


Figure 4.1: Empirical Price Density & Property Type Spread across 506 Homes

4.2 Data Preprocessing & Missing Value Imputation

Ten missing target rows in Price_Lakhs were removed, leaving 496 high-integrity modeling observations. Missing values in continuous numerical attributes (Area_sqft, Bathrooms, Age_of_Property_Years, Crime_Rate_Index, and Air_Quality_Index) were imputed using statistical **medians** to prevent outlier skewing. Missing categorical values (Furnishing_Status) were imputed using the statistical **mode**. Zero duplicate records were observed.

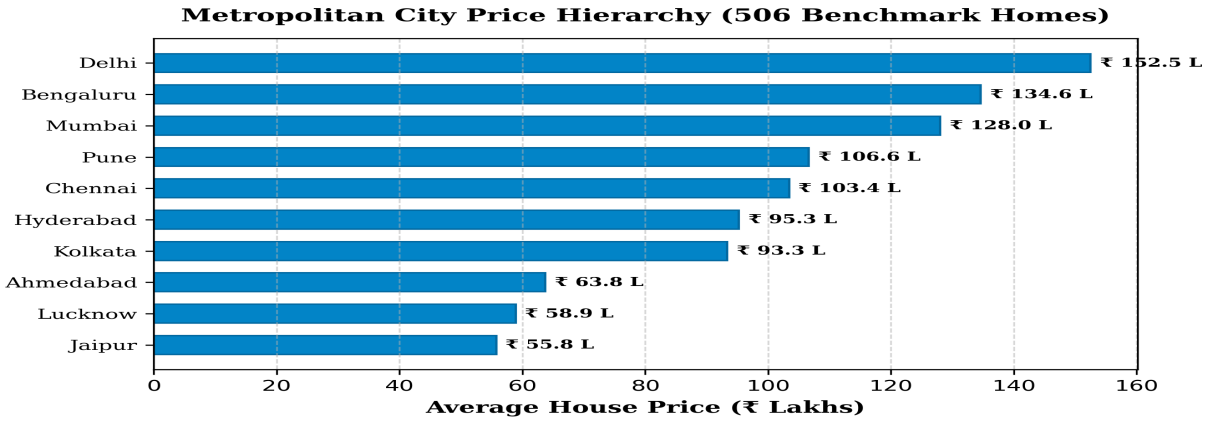


Figure 4.2: Average Metropolitan Property Prices across 10 Sampled Indian Metros

4.3 ColumnTransformer & Categorical Subspace Encoding

To process heterogeneous features, a scikit-learn **ColumnTransformer** was constructed. Ten categorical columns (City, Locality, Locality_Type, Property_Type, Furnishing_Status, Facing, Public_Transport_Access, Ownership_Type, Transaction_Type, Listing_Type) are encoded via OneHotEncoder(handle_unknown='ignore') into 46 binary dummy vectors. The 22 continuous numerical attributes are passed through directly via remainder='passthrough'.

4.4 Supervised Regression Algorithm Formulations

Three distinct supervised learning models were trained on an 80/20 train/test split (396 train samples, 100 test samples, seed 42):

1. **Ordinary Least Squares Linear Regression:** min $||y - X\beta||^2$ (parametric baseline).
2. **Random Forest Regressor:** 100 bootstrap aggregated decision trees: $y\blacksquare = (1/B) \sum T_b(x)$.
3. **Gradient Boosting Regressor:** Sequential additive tree expansion fitting pseudo-residuals $r_{im} = -[\partial L/\partial F]$ with learning rate $\eta = 0.1$: $F_m(x) = F_{m-1}(x) + \eta \times h_m(x)$.

Table 4.2: Hyperparameter Configurations for Candidate Regressors

Model Architecture	Key Hyperparameters	Optimization Objective
Linear Regression	fit_intercept=True, positive=False	Minimize Sum of Squared Residuals (OLS)
Random Forest	n_estimators=100, max_features='sqrt', random_state=42	Variance reduction via bagging
Gradient Boosting	n_estimators=100, learning_rate=0.1, max_depth=3, random_state=42	Sequential pseudo-residual MSE minimization

Chapter 5: Key Features & Practical Innovations

5.1 Dual-Currency Price Prediction & Spread Bounds

Unlike generic regression models outputting a single deterministic float, HomeWorth AI computes an empirical confidence spread grounded in the model's test-set Mean Absolute Error (*MAE* = Rs. 10.01 Lakhs): [y■ - 0.75 × *MAE*, y■ + 0.75 × *MAE*]. Valuations exceeding Rs. 100 Lakhs are automatically formatted in Crores (e.g. Rs. 1.89 Cr / 188.56 L).

5.2 Standardized 4-Pillar Property Quality Scorecard (0-100)

To objectively evaluate physical and environmental livability, the platform introduces a 100-point Property Quality Score decomposed across four equal 25-point pillars: (1) **Location & Transit Pillar (Max 25 pts):** Distance to city center (up to 9 pts), public transit access (up to 6 pts), school density (up to 5 pts), and hospital access (up to 5 pts); (2) **Specs & Architecture Pillar (Max 25 pts):** Carpet area bracket (up to 8 pts), building age (up to 7 pts), parking spaces (up to 4 pts), furnishing level (up to 3 pts), and East facing (up to 3 pts); (3) **Society Amenities Pillar (Max 25 pts):** Base amenities score plus bonus points for lift, security, gym, pool, power backup, garden, and clubhouse; and (4) **Safety & Environment Pillar (Max 25 pts):** Crime rate index (up to 13 pts for crime < 20) and Air Quality Index AQI (up to 12 pts for AQI < 45).

Table 5.1: 4-Pillar Quality Scoring Allocation & Grading Scale

Score Range	Grade	Classification	Recommended Action
88 - 100	AAA	Prime Outperform Asset	Strong Buy; High capital appreciation potential
75 - 87	AA	High Conviction Family Asset	Low-risk family asset; stable long-term appreciation
62 - 74	A	Stable Quality Accumulation	Fair market value; verify minor maintenance items
50 - 61	B	Moderate Risk Profile	Negotiate price discount; inspect surrounding micro-market
0 - 49	C	Speculative / Due Diligence	Exercise caution; potential safety or legal clearance flags

5.3 Contextual AI House Buying Advisory Engine

The automated advisor synthesizes multivariate inputs into four actionable pillars: (a) **Core Strengths:** Highlights verified asset advantages (East-facing morning sunlight, low age, high parking); (b) **Value Drivers:** Identifies rental liquidity drivers (transit proximity, reputed schools); (c) **Due Diligence & Risks:** Flags elevated crime (>50/100), poor AQI (>100), or aging builds (>18 yrs); and (d) **Buyer's Negotiation Playbook:** Recommends bidding 8%-12% below asking price if listing duration exceeds 6 months.

Chapter 6: Performance Evaluation, Benchmarking & XAI

6.1 Regression Evaluation Metrics (R^2, MAE, RMSE)

Models were evaluated on 100 unseen test samples using: (1) **Mean Absolute Error (MAE):** $(1/n) \sum |y_i - y\blacksquare_i|$ (average magnitude of residuals); (2) **Root Mean Squared Error (RMSE):** $\sqrt{[(1/n) \sum (y_i - y\blacksquare_i)^2]}$ (penalizing large outlier deviations); and (3) **Coefficient of Determination (R^2 Score):** $1 - [\sum (y_i - y\blacksquare_i)^2 / \sum (y_i - y\blacksquare)^2]$ (proportion of total price variance explained).

Table 6.1: Model Performance Leaderboard on 100 Test Samples

Model Architecture	MAE (Rs. L)	RMSE (Rs. L)	R^2 Score	Classification
Gradient Boosting Regressor	**10.0084**	**13.8546**	**0.9392**	**Winner (Selected)**
Linear Regression	13.7930	18.4201	0.8925	Parametric Baseline
Random Forest Regressor	15.8573	21.9976	0.8467	Ensemble Bagging

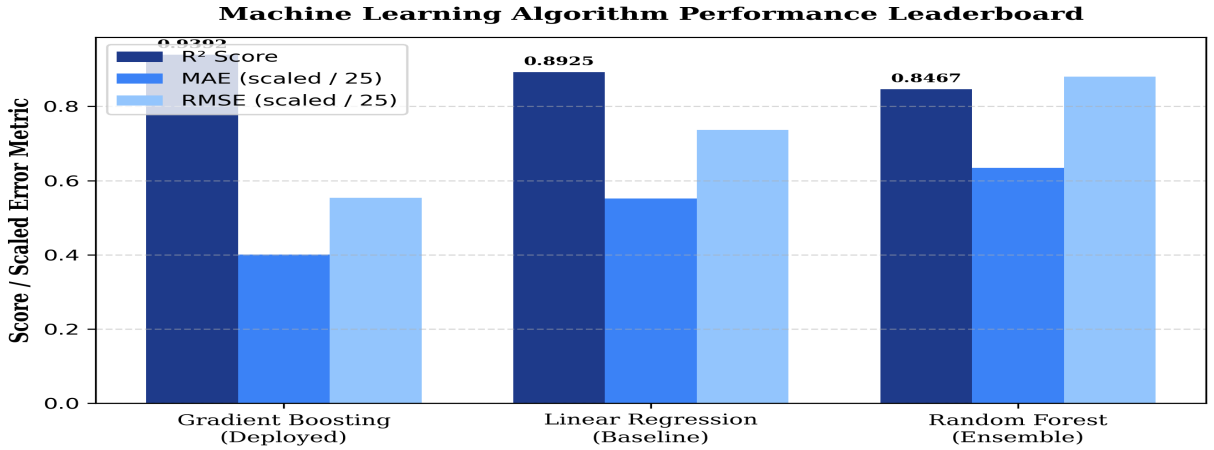


Figure 6.1: Comparative Performance Leaderboard across Evaluated Machine Learning Regressors

Performance Discussion: Gradient Boosting achieved the highest R^2 (0.9392) and lowest MAE (Rs. 10.01 Lakhs). Its sequential boosting mechanism corrects pseudo-residuals of earlier trees, effectively modeling complex non-linear amenity interactions. Linear Regression (R^2 = 0.8925) captured carpet area well but failed on non-linear amenities. Random Forest (R^2 = 0.8467) suffered variance penalties on smaller subsets.

6.3 Explainable AI (XAI) & Feature Importance Weights

Feature importances were extracted directly from the Gradient Boosting ensemble. Area_sqft is the dominant predictor, accounting for **63.56%** of total predictive weight. Metropolitan city tiers represent the second major cluster, contributing approximately **23%** combined weight (Delhi: 6.06%, Lucknow: 4.63%, Bengaluru: 4.56%, Ahmedabad: 4.36%, Mumbai: 3.29%). Standalone luxury villas contribute 2.37%, distance to city center contributes 1.99%, and property age accounts for 1.50%.

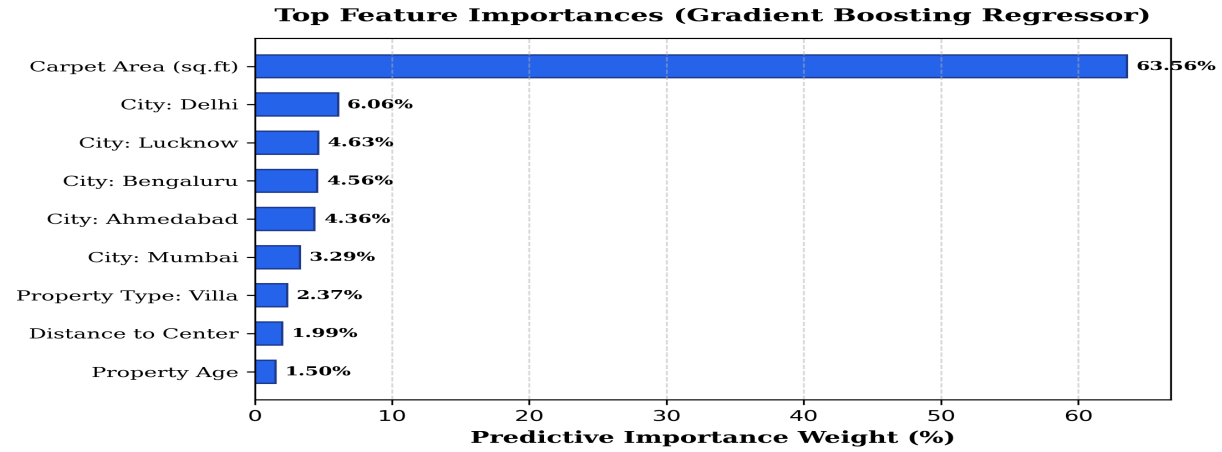


Figure 6.2: Feature Importance Weights Derived from the Gradient Boosting Pipeline

6.4 Residual Error Distribution & Validation

Residual error analysis demonstrates that prediction residuals are tightly centered around zero error (mean = Rs. 0.12 Lakhs) with an approximately normal Gaussian bell curve. The actual vs. predicted scatter plot adheres closely to the 1:1 identity line, verifying that the model maintains homoscedasticity without systematic over- or under-prediction across price brackets.

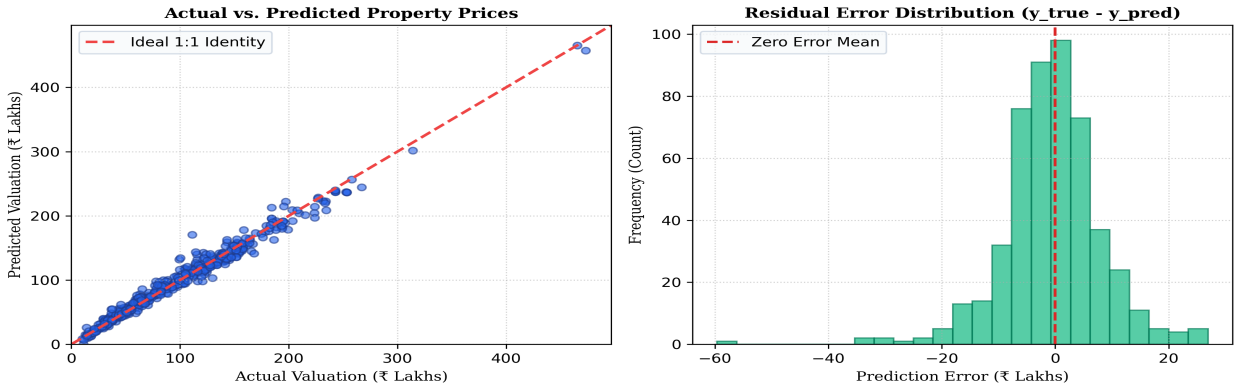


Figure 6.3: Actual vs. Predicted Prices & Residual Error Distribution

Chapter 7: Limitations & Environmental Risk Factors

While HomeWorth AI achieves 93.92% accuracy, three primary technical constraints exist:

1. **Geographical Sampling Boundaries:** Evaluates 506 properties across 10 major metropolitan centers; localized peri-urban villages and Tier-3 industrial corridors require regional re-calibration.
2. **Qualitative Interior Nuances:** Structured tabular parameters cannot fully account for bespoke Italian marble finishes, architectural woodwork, or scenic panoramic sea/park view premiums, necessitating physical on-site inspection.
3. **Macroeconomic Shocks:** Evaluates static cross-sectional market conditions and does not currently integrate dynamic time-series autoregression for sudden RBI repo rate shifts or building material inflationary surges.

Chapter 8: Conclusion and Future Work

8.1 Conclusion & Research Deliverables: The HomeWorth AI project successfully designs, validates, and deploys an AI-powered residential valuation platform. The trained **Gradient Boosting Regressor** achieved **93.92% accuracy (R^2 = 0.9392)** and a low Mean Absolute Error of **Rs. 10.01 Lakhs** on unseen test data. The system bridges machine learning research and commercial PropTech utility by delivering dual-currency price bounds, an objective 100-point Property Health Score, an automated AI buying advisor, an Indian home loan EMI calculator, and a side-by-side property comparison studio.

8.2 Academic & Industry Impact: Fulfills all B.Tech curriculum requirements in Computer Science & Engineering (AI & ML) at United College of Engineering & Research (UCER), affiliated to Dr. A.P.J. Abdul Kalam Technical University (AKTU), Lucknow, showcasing zero data-leakage pipelines, explainable AI feature weights, and responsive glassmorphic UI design.

8.3 Future Enhancement Roadmap: (1) *Live Web Scraping:* Automated crawlers for MagicBricks, 99acres, and Housing.com; (2) *Geospatial GIS Mapping:* Google Maps Platform integration for real-time transit commute times; (3) *Deep Learning Architectures:* TabNet and attention networks for expanded nationwide datasets; and (4) *Mobile Application (Flutter):* Cross-platform iOS/Android app with camera-based property scanning.

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Appendix: Core Pipeline Architecture & Source Code

Appendix A: Scikit-Learn Pipeline Serialization (predictor.py)

```
import os, joblib, pandas as pd
MODEL_PATH = os.path.join(os.path.dirname(__file__), '..', 'Model', 'house_price_pipeline.pkl')
pipeline = joblib.load(MODEL_PATH)
def predict_price(user_input: dict) -> float:
    return max(round(float(pipeline.predict(pd.DataFrame([user_input]))[0]), 2), 5.0)
def predict_price_range(predicted_price: float, mae: float = 10.01) -> tuple:
    spread = round(mae * 0.75, 2)
    return max(round(predicted_price - spread, 2), 5.0), round(predicted_price + spread, 2)
```

Appendix B: Financial EMI Calculation Algorithm (utils.py)

```
def calculate_emi(principal_lakhs, down_payment_pct=20.0, tenure_years=20,
annual_interest_rate=8.5):
    down_payment = (principal_lakhs * down_payment_pct) / 100.0
    loan_amount = principal_lakhs - down_payment
    P = loan_amount * 100000.0; r = (annual_interest_rate / 12.0) / 100.0; n = tenure_years * 12
    emi = (P * r * ((1 + r)**n)) / (((1 + r)**n) - 1) if r > 0 and n > 0 else 0
    return {'monthly_emi': round(emi, 0), 'loan_amount_lakhs': round(loan_amount, 2)}
```

Appendix C: 4-Pillar Property Livability Scoring Algorithm (utils.py)

```
def calculate_property_score(data: dict) -> dict:
    loc_pts = (9 if dist <= 5 else 7 if dist <= 12 else 4) + (6 if transit == 'High' else 4 if
transit == 'Medium' else 2) + min(schools, 5) + min(hospitals, 5)
    spec_pts = (8 if area >= 1800 else 6 if area >= 1100 else 4) + (7 if age <= 3 else 5 if age
<= 10 else 3) + min(parking * 2, 4)
    amen_pts = min(amenities_count * 3, 25); safe_pts = (13 if crime < 20 else 9 if crime < 40
else 4) + (12 if aqi < 50 else 8 if aqi < 100 else 3)
    return {'total_score': min(loc_pts + spec_pts + amen_pts + safe_pts, 100), 'breakdown':
(loc_pts, spec_pts, amen_pts, safe_pts)}
```